

22S Series — Metal Oxide Varistors

22 mm square, epoxy-encapsulated MOVs for surge protective devices · Datasheet · Rev. 2026-10-21

Description

The 22S series is a line of 22 mm square, epoxy-encapsulated zinc oxide varistors made by Guiyang High-Tech Yige Electronic Co., Ltd. for surge protective device (SPD) manufacturers and OEMs. Rated In 10 kA / Imax 20 kA (8/20 µs), the range covers varistor voltages from 100 V to 820 V in 22 model grades. The varistor discs are pressed, sintered, metallized, coated and tested in our own Guiyang plant — not bought in — which is how tolerances and surge ratings stay consistent from batch to batch. In North America, the series is supported by Flyboat Electronic LLC, the factory's sales and engineering center in Bellevue, WA.



Features

- Centre melt-through under power-frequency overload: the melt-through (puncture) point is engineered to occur at the centre of the disc, underneath the copper electrode, so heat is absorbed and conducted straight to the disconnection point — the chip resists melt-through longer and a mating thermal disconnector trips faster.
- 22 mm square MOVs, epoxy encapsulated, for SPD manufacturers and OEMs
- 22 UL-recognized model grades — varistor voltage from 100 V to 820 V
- UL 1449 Recognized, File E335803 (VZCA2 US / VZCA8 Canada)
- In 10 kA, Imax 20 kA (8/20 µs)
- Low-residual-voltage range available for tighter clamping
- Standard 22 mm square footprint used across the SPD industry

Agency approvals

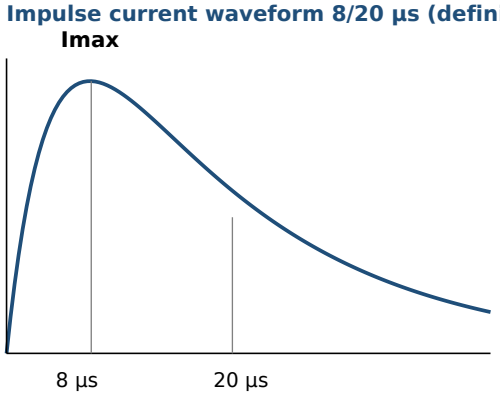
Standard	Category	File / status
UL 1449	VZCA2 (US) — surge protective components	File E335803, UL Recognized Component
CSA C22.2 No. 269.5-17	VZCA8 (Canada)	File E335803, UL Recognized Component

22 model grades are recognized, covering varistor voltages from 100 V to 820 V (model codes 101–821). Verify in UL Product iQ (productiq.ulprospector.com) by searching file number E335803 — the record lists exactly which model codes are covered.

Absolute maximum ratings

Parameter	Range / limit	Unit
Continuous operating voltage, AC (VM(AC)rms)	60 to 510	V
Continuous operating voltage, DC (VM(DC))	85 to 670	V
Nominal discharge current In (8/20 µs)	10	kA
Maximum discharge current Imax (8/20 µs)	up to 20	kA
Maximum energy absorption (2 ms)	up to 730	J

Impulse current waveform 8/20 µs (definition)



CAUTION: stresses above the ratings in the tables below may cause permanent damage. The ranges above summarize the per-model ratings in the sections below; always design to the row for the exact model code.

Model naming and standard marking

Model naming: 22S K 431 — 22 = nominal size (mm), S = square; K = varistor-voltage tolerance ±10 % (J = ±5 %); 431 = varistor-voltage code (first two digits significant, last digit = number of zeros; 431 = 430 V). Further suffixes give the lead/structure form; -U = uncoated bare electrode disc (no mark = epoxy-coated by default). Standard marking: the manufacturer logo and the model code are etched on the encapsulation.

1. Electrical characteristics — standard ratings

U1mA = varistor voltage at DC 1 mA; ΔU1mA = tolerance; UAC / UDC = maximum continuous operating voltage; In = nominal discharge current (8/20 µs); Imax = maximum discharge current (8/20 µs); UP = clamping voltage at In (reference value–ratio); ETM = maximum energy absorption (2 ms).

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
22SK101	100	±10	60	85	10	20	360-3.6	100	✓	✓	—
22SK121	120	±10	75	100	10	20	400-3.6	125	✓	✓	—
22SK151	150	±10	95	125	10	20	450-3.0	160	✓	✓	—
22SK181	180	±10	115	150	10	20	490-2.8	190	✓	✓	—
22SK201	200	±10	130	170	10	20	520-2.6	220	✓	✓	—
22SK221	220	±10	140	180	10	20	550-2.6	240	✓	✓	—
22SK241	240	±10	150	200	10	20	600-2.6	260	✓	✓	—
22SK271	270	±10	175	225	10	20	680-2.6	300	✓	✓	—
22SK301	300	±10	195	250	10	20	770-2.6	330	✓	✓	—
22SK331	330	±10	210	270	10	20	830-2.55	360	✓	✓	—
22SK361	360	±10	230	300	10	20	910-2.55	410	✓	✓	—
22SK391	390	±10	250	320	10	20	1000-2.55	450	✓	✓	—
22SK431	430	±10	275	350	10	20	1050-2.45	510	✓	✓	—
22SK471	470	±10	300	385	10	20	1100-2.4	550	✓	✓	—
22SK511	510	±10	320	410	10	20	1150-2.35	590	✓	✓	—
22SK561	560	±10	350	450	10	20	1250-2.35	640	✓	✓	—
22SK621	620	±10	385	505	10	20	1450-2.3	700	✓	✓	—
22SK681	680	±10	420	560	10	20	1600-2.3	710	✓	✓	—
22SK711	710	±10	440	585	10	20	1650-2.3	715	✓	✓	—
22SK751	750	±10	460	615	10	20	1700-2.3	720	✓	✓	—
22SK781	780	±10	485	640	10	20	1750-2.3	725	✓	✓	—
22SK821	820	±10	510	670	10	20	1800-2.3	730	✓	✓	—

2. Electrical characteristics — low-residual-voltage ratings

Model	U1mA (V)	ΔU1mA (%)	UAC (V)	UDC (V)	In (kA)	Imax (kA)	UP (V)	ETM (J)	UL	C-UL	TÜV
22SK431	430	±10	275	350	10	20	990-2.2	510	✓	✓	—
22SK471	470	±10	300	385	10	20	1070-2.2	550	✓	✓	—
22SK511	510	±10	320	410	10	20	1150-2.2	590	✓	✓	—
22SK561	560	±10	350	450	10	20	1250-2.15	640	✓	✓	—
22SK621	620	±10	385	505	10	20	1350-2.1	700	✓	✓	—
22SK681	680	±10	420	560	10	20	1520-2.1	710	✓	✓	—
22SK711	710	±10	440	585	10	20	1600-2.1	715	✓	✓	—
22SK751	750	±10	460	615	10	20	1700-2.1	720	✓	✓	—
22SK781	780	±10	485	640	10	20	1760-2.1	725	✓	✓	—
22SK821	820	±10	510	670	10	20	1800-2.08	730	✓	✓	—

Dimensions and thickness

Encapsulated thickness Tmax (mm), standard & low-clamping ranges — 101: 3.8 · 121: 3.8 · 151: 3.8 · 181: 3.9 · 201: 4.0 · 221: 4.1 · 241: 4.2 · 271: 4.4 · 301: 4.6 · 331: 4.7 · 361: 4.9 · 391: 5.1 · 431: 5.2 · 471: 5.5 · 511: 5.7 · 561: 6.0 · 621: 6.3 · 681: 6.3 · 711: 6.6 · 751: 7.1 · 821: 7.2.
Dimension drawings for each package form are available on request.

Ordering and support: samples and design-in support — North America: Thomas Dong, Flyboat Electronic LLC, Bellevue, WA (thomasd@flyboatmov.com); China: Colin Fei (colinf@flyboatmov.com). Specifications are subject to change without notice; verify critical ratings against the model row above before design freeze.